

INTER-REGION VPC PEERING

A step-by-step hands-on configuration guide for connecting VPCs across AWS Regions using VPC Peering.

REGIONS

Mumbai & Singapore

DOCUMENT TYPE

Lab Guide

VPC Peering

Route Tables

Security Groups

EC2 Connectivity Test

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INTER-REGION VPC PEERING:

STEP 01

Create VPC in Mumbai

1. AWS Console → **VPC**
2. Select Region: **Mumbai**
3. Click **Create VPC**
4. Name: **Mumbai-VPC**
5. IPv4 CIDR: **10.10.0.0/16**
6. Click **Create VPC**

The screenshot displays the AWS Management Console interface for creating a new VPC. The breadcrumb navigation shows 'VPC > Your VPCs > Create VPC'. The page title is 'Create VPC' with an 'Info' link. A descriptive text states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.'

The 'VPC settings' section contains the following fields and options:

- Resources to create** (Info link): A radio button is selected for 'VPC only', and another option 'VPC and more' is available.
- Name tag - optional** (Info link): A text input field contains the value 'vpc1'. A note below states: 'Creates a tag with a key of 'Name' and a value that you specify.'
- IPv4 CIDR block** (Info link): Two radio buttons are present. 'IPv4 CIDR manual input' is selected, and 'IPAM-allocated IPv4 CIDR block' is the other option.
- IPv4 CIDR**: A text input field contains the value '10.10.0.0/16'. A note below states: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block** (Info link): Three radio buttons are present. 'No IPv6 CIDR block' is selected, and the other two options are 'IPAM-allocated IPv6 CIDR block' and 'Amazon-provided IPv6 CIDR block'.

STEP 02

Create VPC in Singapore

1. Change Region → Singapore
2. Go to **VPC** → **Your VPCs**
3. Click **Create VPC**
4. Name: Singapore-VPC
5. IPv4 CIDR: 10.20.0.0/16
6. Click **Create VPC**

VPC CIDRs must not overlap.

   [Alt+S]     Asia Pacific (Mumbai) Shreya

[VPC](#) > [Subnets](#) > Create subnet

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-087000c63ce317a97 (vpc1) ▼

Associated VPC CIDRs

IPv4 CIDRs
10.10.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

vpc1-sub-public

The name can be up to 256 characters long.

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-087000c63ce317a97 (vpc1) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.10.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

vpc1-sub-private

The name can be up to 256 characters long.

STEP 03

Mumbai Internet Gateway

The screenshot shows the AWS Management Console interface for creating an internet gateway. The breadcrumb navigation is VPC > Internet gateways > Create internet gateway. The page title is 'Create internet gateway' with an 'Info' link. A description states: 'An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.' Under 'Internet gateway settings', the 'Name tag' section has a text input field containing 'Mumbai-IGW'. The 'Tags - optional' section explains that a tag is a label for an AWS resource and shows a table with one tag: Key 'Name' and Value 'Mumbai-IGW'. There is an 'Add new tag' button and a note 'You can add 49 more tags.' At the bottom right are 'Cancel' and 'Create internet gateway' buttons.

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Shreya_Panchal

VPC > Internet gateways > Create internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Mumbai-IGW

Tags - *optional*

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - <i>optional</i>
Name	Mumbai-IGW

[Add new tag](#) You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

The screenshot shows the AWS Management Console interface for attaching an internet gateway to a VPC. The breadcrumb navigation is VPC > Internet gateways > Attach to VPC (igw-051ddc476a1ad1d57). The page title is 'Attach to VPC (igw-051ddc476a1ad1d57)' with an 'Info' link. A description states: 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' Under 'Available VPCs', there is a text input field containing 'vpc-087000c63ce317a97'. At the bottom right are 'Cancel' and 'Attach internet gateway' buttons.

VPC > Internet gateways > Attach to VPC (igw-051ddc476a1ad1d57)

Attach to VPC (igw-051ddc476a1ad1d57) [Info](#)

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

vpc-087000c63ce317a97

[AWS Command Line Interface command](#)

[Cancel](#) [Attach internet gateway](#)

STEP 04

Mumbai Route Table

aws Search [Alt+S] Asia Pacific (Mumbai) Shreya_Panchal (673034950588) root

VPC > Route tables > Create route table

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Mumbai-RT

VPC
The VPC to use for this route table.
vpc-087000c63ce317a97 (vpc1)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q Mumbai-RT X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

aws Search [Alt+S] Asia Pacific (Mumbai) Shreya_Panchal (673034950588) root

VPC > Route tables > rtb-0c07dccc0b169068 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	vpc1-sub-public	subnet-0ab5287345e15667b	10.10.1.0/24	-	Main (rtb-01b34c81199fd69d7)

Selected subnets

subnet-0ab5287345e15667b / vpc1-sub-public X

Cancel Save associations

aws Search [Alt+S] Asia Pacific (Mumbai) Shreya_Panchal (673034950588) root

VPC > Route tables > rtb-0c07dccc0b169068 > Edit routes

Edit routes: rtb-0c07dccc0b169068 / Mumbai-RT

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0 X	Internet Gateway	-	No	CreateRoute
	igw-051ddc476a1ad1d57			

Add route

Cancel Preview Save changes

STEP 05

Singapore me VPC create

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

Singapore-VPC

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.20.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block

STEP 06

Singapore subnet

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-0e27f833eb6507197 (Singapore-VPC)

Associated VPC CIDRs

IPv4 CIDRs
10.20.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

vpc2-sub-private
The name can be up to 256 characters long.

STEP 07

Singapore Internet Gateway

The screenshot shows the 'Create internet gateway' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Internet gateways > Create internet gateway'. The page title is 'Create internet gateway' with an 'Info' icon. A description states: 'An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.' The 'Internet gateway settings' section includes a 'Name tag' field with the value 'Singapore-IGW'. The 'Tags - optional' section shows a table with one tag: 'Name' with value 'Singapore-IGW'. There are 'Add new tag' and 'Remove' buttons. At the bottom right are 'Cancel' and 'Create internet gateway' buttons.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Singapore-IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional
Name	Singapore-IGW

[Add new tag](#) [Remove](#)

You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

The screenshot shows the 'Attach to VPC' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Internet gateways > Attach to VPC (igw-06f153661142a8d74)'. A green notification banner at the top states: 'The following internet gateway was created: igw-06f153661142a8d74 - Singapore-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet.' The page title is 'Attach to VPC (igw-06f153661142a8d74)' with an 'Info' icon. The 'VPC' section includes a description: 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' The 'Available VPCs' section shows a table with one VPC: 'vpc-0e27f833eb6507197' with the name 'Singapore-VPC'. There are 'Attach to a VPC' and 'Cancel' buttons.

Attach to VPC (igw-06f153661142a8d74) [Info](#)

The following internet gateway was created: igw-06f153661142a8d74 - Singapore-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-0e27f833eb6507197	Singapore-VPC
-----------------------	---------------

[Attach to a VPC](#) [Cancel](#) [Attach internet gateway](#)

STEP 08

Singapore Route Table

The screenshot shows the 'Create route table' page in the AWS console. The breadcrumb navigation is 'VPC > Route tables > Create route table'. The page title is 'Create route table' with an 'Info' icon. A description states: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.'

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Singapore-RT

VPC
The VPC to use for this route table.
vpc-0e27f833eb6507197 (Singapore-VPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Singapore-RT

Value - optional
Singapore-RT

Add new tag
You can add 49 more tags.

Buttons: Cancel, Create route table

The screenshot shows the 'Edit subnet associations' page in the AWS console. The breadcrumb navigation is 'VPC > Route tables > rtb-086f11dfe033fcb37 > Edit subnet associations'. The page title is 'Edit subnet associations'. A description states: 'Change which subnets are associated with this route table.'

Available subnets (1/1)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ vpc2-sub-private	subnet-003e2bb580329b959	10.20.1.0/24	-	Main (rtb-04dbbf074948321b4)

Selected subnets

subnet-003e2bb580329b959 / vpc2-sub-private

Buttons: Cancel, Save associations

The screenshot shows the 'Edit routes' page in the AWS console. The breadcrumb navigation is 'VPC > Route tables > rtb-086f11dfe033fcb37 > Edit routes'. The page title is 'Edit routes: rtb-086f11dfe033fcb37 / Singapore-RT'.

Destination	Target	Status	Propagated	Route Origin
10.20.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

STEP 09

EC2 launch Singapore EC2

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
singapore-ec2-private [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian [Browse more AMIs](#)
Including AMIs from AWS, Marketplace and other providers

▼ **Summary**

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-0532913178263be11

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*
admin [Create new key pair](#)

▼ **Network settings** [Info](#)

VPC - *required* [Info](#)
vpc-0e27f833eb6507197 (Singapore-VPC)
10.20.0.0/16

Subnet [Info](#)
subnet-00362bb580329b959 vpc2-sub-private [Create new subnet](#)
VPC: vpc-0e27f833eb6507197 Owner: 673034950588
Availability Zone: ap-southeast-1a (ap-s1-ez2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.20.1.0/24

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

▼ **Summary**

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-0532913178263be11

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

Mumbai EC2

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

mumbai-ec2-public

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

CancelLaunch instancePreview code

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

admin@123

Create new key pair

Network settings

VPC - required

vpc-087000c63ce317a97 (Mumbai-vpc)

10.10.0.0/16

Subnet

subnet-0ab5287345e15667b

vpc1-sub-public

Create new subnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security groupSelect existing security group

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

CancelLaunch instancePreview code

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STEP 10

Create VPC Peering

1. Change Region → **Mumbai**
2. Go to **VPC** → **Peering connections**
3. Click **Create peering connection**
4. Name: **Mumbai-Singapore-Peering**
5. Requester VPC: **Mumbai-VPC**
6. Select **Another Region**
7. Select Region: **Singapore**
8. Select Accepter VPC: **Singapore-VPC**
9. Click **Create peering connection**

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Mumbai-Singapore-Peering

Select a local VPC to peer with

VPC ID (Requester)
vpc-087000c63ce317a97 (Mumbai-vpc)

VPC CIDRs for vpc-087000c63ce317a97 (Mumbai-vpc)

CIDR	Status	Status reason
10.10.0.0/16	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☐ This Region (ap-south-1)
☒ Another Region
 Asia Pacific (Singapore) (ap-southeast-1)

VPC ID (Accepter)
vpc-0e27f833eb6507197

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
Mumbai-Singapore-Peering

[Add new tag](#)

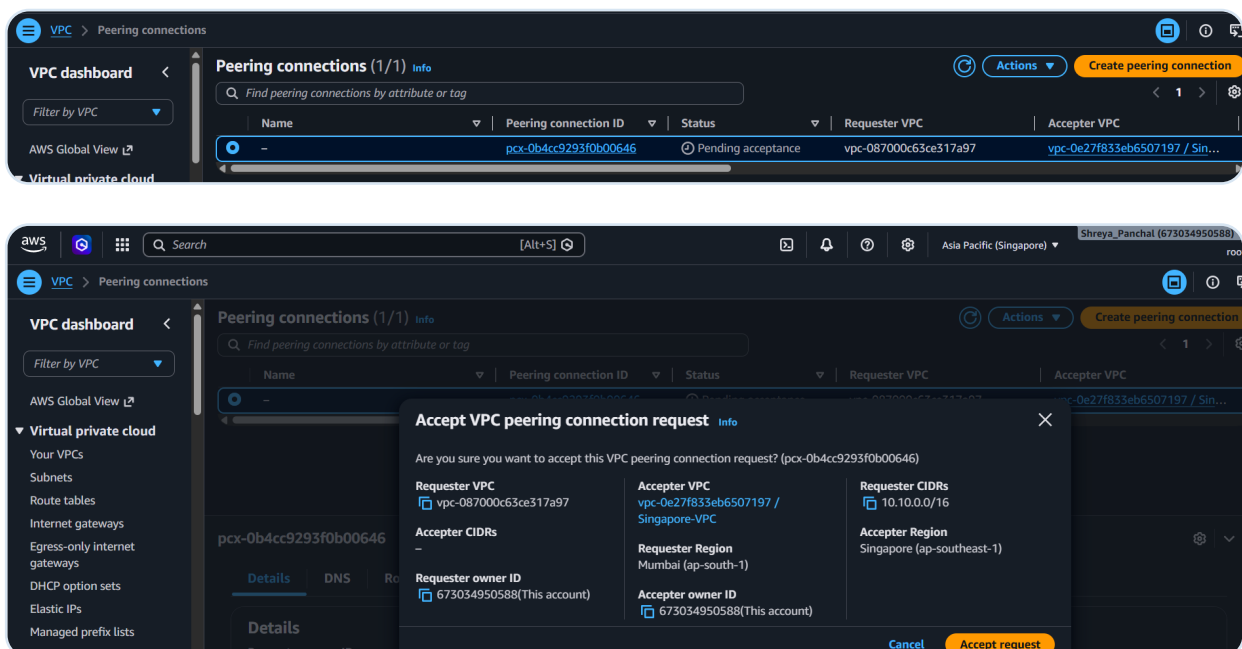
You can add 49 more tags.

[Cancel](#) [Create peering connection](#)

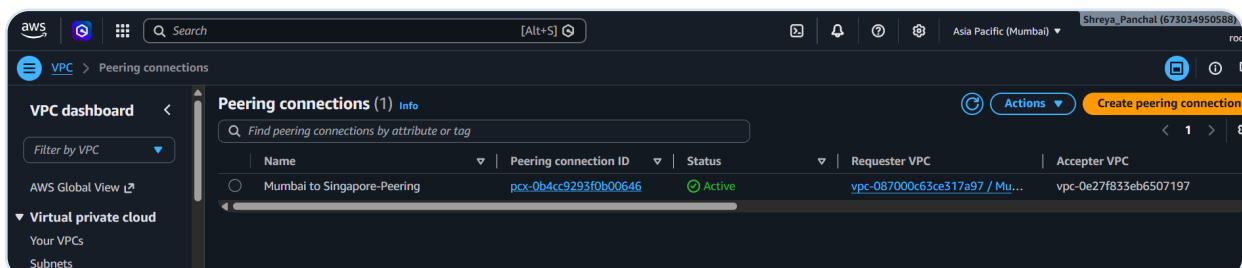
STEP 11 Accept Peering Request

1. Change Region → **Singapore**
2. Go to **VPC** → **Peering connections**
3. Select the pending peering connection
4. Click **Actions**
5. Select **Accept request**
6. Confirm

Status should become: **Active**

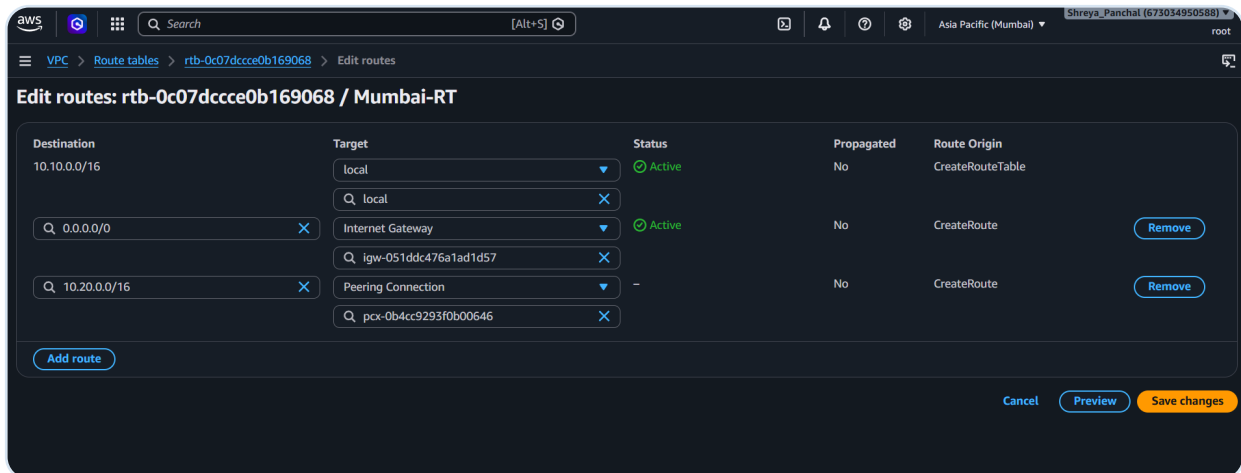


STEP 12 Mumbai Route Table me peering route

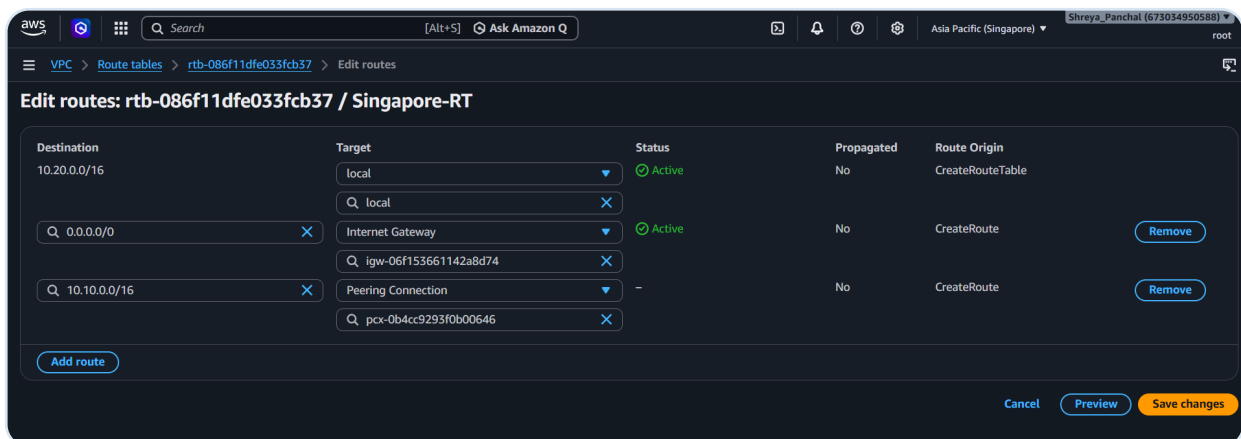


STEP 13

Mumbai Route Table me reverse route



Singapore Route Table me reverse route



STEP 13

Configure Security Groups

Security Group	Rule	Type	Source
Mumbai EC2 Security Group	Allow traffic from Singapore VPC	All ICMP - IPv4	10.20.0.0/16
Singapore EC2 Security Group	Allow traffic from Mumbai VPC	All ICMP - IPv4	10.10.0.0/16

STEP 14

Test Connectivity

From Mumbai EC2:

```
ping <Singapore-EC2-Private-IP>
```

```
ubuntu@ip-10-10-1-188: ~  
C:\Users\Shreya>cd Downloads  
C:\Users\Shreya\Downloads>  
C:\Users\Shreya\Downloads>ping 13.234.240.135  
  
Pinging 13.234.240.135 with 32 bytes of data:  
Reply from 13.234.240.135: bytes=32 time=19ms TTL=58  
Reply from 13.234.240.135: bytes=32 time=18ms TTL=58  
Reply from 13.234.240.135: bytes=32 time=18ms TTL=58  
  
Ping statistics for 13.234.240.135:  
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 18ms, Maximum = 19ms, Average = 18ms  
Control-C  
^C  
C:\Users\Shreya\Downloads>ssh -i "admin@123.pem" ubuntu@13.234.240.135  
The authenticity of host '13.234.240.135 (13.234.240.135)' can't be established.  
ED25519 key fingerprint is SHA256:0VtsjtUdNfs44UbU8xn0spanmMZ56jhLksio255Fte8.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '13.234.240.135' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat Aug  8 05:05:21 UTC 2026  
  
System load:  0.0           Temperature:    -273.1 C  
Usage of /:   30.5% of 6.61GB Processes:      116  
Memory usage: 26%          Users logged in: 0  
Swap usage:   0%           IPv4 address for ens5: 10.10.1.188  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

```
ubuntu@ip-10-10-1-188:~$
ubuntu@ip-10-10-1-188:~$
ubuntu@ip-10-10-1-188:~$ ping 10.20.1.34
PING 10.20.1.34 (10.20.1.34) 56(84) bytes of data.
64 bytes from 10.20.1.34: icmp_seq=1 ttl=64 time=60.5 ms
64 bytes from 10.20.1.34: icmp_seq=2 ttl=64 time=60.5 ms
^C
--- 10.20.1.34 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 60.470/60.475/60.480/0.005 ms
ubuntu@ip-10-10-1-188:~$ |
```

Final Configuration:



Required Steps

Create VPCs → Create Peering → Accept Request → Update Both Route Tables → Configure Security Groups → Ping Test

Expected Result

Mumbai EC2 — Private IP —> Singapore EC2

Inter-Region VPC Peering: Successfully Configured